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ArduSimple: male headers soldered underneath; power switch DOWN (5V IN, IOREF N/C); --role rover (UART2 off).

Feather on 8.5 mm stacking headers (housings on the display side), tails into J5/J6; display up, USB-C at the front. LIS3DH runs from VSYS (its own 3.3 V regulator).

bq24074 optional: fitted -> JP1 open, battery feeds the charger, charger OUT feeds VSYS.

No bq24074 -> fit the JP1 shunt (BATT to VSYS). Never both: the shunt would bypass the charger.

~CHG and ~PGOOD are open drain to Feather D11 / D12 (read with INPUT_PULLUP); the firmware does not use them yet.

Optional, unused by the firmware so far: TIMEPULSE -> A1, EXTINT <- A2, bq ~CE <- A3 (high stops charging; 10k pull-down on the bq), LIS3DH INT2 -> A4.

GPS_EN: Feather A0 to pad TP2; wire TP2 to the S9V11F5 EN pad. Build with -DGPS_POWER_PIN=18 to cut the GPS and XBee in deep sleep.

J12: short EN to GND to switch the Feather off (the regulator and ArduSimple stay on battery power).

Feather S3 Reverse TFT, ArduSimple Budget, LIS3DH, S9V11F5, optional bq24074

race-tracker

Sheet: /

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Title: ESP32 rover carrier

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